

## § 23. The subdeterminants in the wider sense

We can now generalize the considerations of the preceding paragraph in the following way. Just as there we started from the problem of seeking out all terms in the developed determinant  $A$  which contain the factor  $a_1^{(1)}$ , so now we seek all terms which contain the factor

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)},$$

where  $\nu$  may be any number not exceeding  $n$ . These terms are obtained from the principal term

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} a_{\nu+1}^{(\nu+1)} \cdots a_n^{(n)}$$

if, in permuting the lower indices, one leaves  $1, 2, \dots, \nu$  unchanged and permutes only  $\nu + 1, \dots, n$  in all possible ways, with observance of the rule of signs. Thus the aggregate of the required terms is

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \vdots & & \vdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}. \quad (1)$$

I. The determinant of  $n - \nu$  rows which occurs here as a factor, and which we denote by

$$A_{1,2,\dots,\nu}^{1,2,\dots,\nu},$$

arises from  $A$  by suppressing the first  $\nu$  rows and columns.

We shall generalize this result as follows. Choose any  $\nu$  elements

$$a_{\alpha_1}^{(\beta_1)}, a_{\alpha_2}^{(\beta_2)}, \dots, a_{\alpha_\nu}^{(\beta_\nu)} \quad (3)$$

so that no two of them occur in the same row or in the same column, that is, so that neither a lower nor an upper index is repeated. We denote by

$$A_{\alpha_1, \alpha_2, \dots, \alpha_\nu}^{\beta_1, \beta_2, \dots, \beta_\nu} \quad (2)$$

the aggregate of those terms of the determinant which contain the product of these elements as a factor.

By interchanging rows and columns, whereby at most the sign of the determinant is changed, one can always bring the elements (3) into the places of

$$a_1^{(1)}, a_2^{(2)}, \dots, a_\nu^{(\nu)}.$$

Then rule I applies, and one obtains:

II. Apart from sign,  $A_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu}$  is obtained as an  $(n - \nu)$ -rowed determinant by deleting from  $A$  all rows and all columns which meet in one of the elements (3), and by leaving the remaining rows and columns in their order.

For the determination of the sign one uses the following rule. Arrange the lower and upper indices  $1, 2, \dots, n$  in the orders

$$\alpha_1, \alpha_2, \dots, \alpha_\nu, \alpha_{\nu+1}, \dots, \alpha_n, \quad (4)$$

$$\beta_1, \beta_2, \dots, \beta_\nu, \beta_{\nu+1}, \dots, \beta_n, \quad (5)$$

where the missing  $\alpha_{\nu+1}, \dots, \alpha_n$ , and likewise  $\beta_{\nu+1}, \dots, \beta_n$ , are taken in increasing order.

III. The determinant of  $n - \nu$  rows described in II receives the positive or negative sign according as the two arrangements (4), (5) of the digits  $1, 2, \dots, n$  both belong to the same kind or to different kinds.

Indeed, the determinant changes its sign through each interchange of two lower or two upper indices. In order to reduce the general case (2) to the special case (1), one has to make just so many transpositions of upper and lower indices that the permutations (4) and (5) both pass into the original arrangement  $1, 2, 3, \dots, n$ ; the same number of changes of sign has then taken place.

The quantities so defined are called the  $\nu$ -th subdeterminants, or subdeterminants of the  $\nu$ -th order. They are represented by determinants of  $n - \nu$  rows. From III follows the theorem:

IV. The subdeterminant

$$A_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu}$$

changes only its sign if two of its lower or two of its upper indices are interchanged; more generally, it remains unaltered in absolute value if the order of the indices is replaced by any other order, and changes sign or not according as this permutation is of the second or of the first kind.

If  $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_\nu$  denotes any ordering of  $\alpha_1, \alpha_2, \dots, \alpha_\nu$ , then the determinant  $A$  also contains the complex of terms

$$\pm a_{\bar{\alpha}_1}^{(\beta_1)} a_{\bar{\alpha}_2}^{(\beta_2)} \dots a_{\bar{\alpha}_\nu}^{(\beta_\nu)} A_{\bar{\alpha}_1, \dots, \bar{\alpha}_\nu}^{\beta_1, \dots, \beta_\nu}.$$

Collecting all these terms gives the complex

$$A_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu} \sum \pm a_{\alpha_1}^{(\beta_1)} a_{\alpha_2}^{(\beta_2)} \dots a_{\alpha_\nu}^{(\beta_\nu)}. \quad (6)$$

The  $\nu$ -rowed determinant which occurs in (6) is called the subdeterminant complementary to  $A_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu}$ , and is denoted by

$$B_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu}.$$

It contains exactly the rows and columns missing in  $A_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu}$ , and, apart from sign, agrees with a subdeterminant of order  $n - \nu$ . Thus the complex of terms (6) is denoted by

$$A_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu} B_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu}. \quad (7)$$

If now, for  $\alpha_1, \dots, \alpha_\nu$ , one chooses each combination of  $\nu$  of the digits  $1, 2, \dots, n$ , while  $\beta_1, \dots, \beta_\nu$  is held fixed, one obtains the same number of complexes of the form (7), and every term of the determinant  $A$  occurs in one and only one of them.

V. Consequently, by summing all expressions (7), one obtains the determinant  $A$ :

$$A = \sum A_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu} B_{\alpha_1, \dots, \alpha_\nu}^{\beta_1, \dots, \beta_\nu}. \quad (8)$$

Of course one may also keep the combination of the  $\alpha$ 's fixed and sum with respect to the  $\beta$ 's. This is Laplace's theorem.

Another representation of the determinant  $A$  by the first and second subdeterminants is obtained as follows. Choose in  $A$  any two rows which intersect in an element, say  $a_\nu^{(\mu)}$ . In every term of  $A$  there occurs one element with lower index  $\nu$  and one with upper index  $\mu$ . Hence one may write

$$A = a_\nu^{(\mu)} A_\nu^{(\mu)} + \sum a_\nu^{(i)} a_k^{(\mu)} A_{\nu k}^{\mu i}, \quad (9)$$

or, according to IV,

$$A = a_\nu^{(\mu)} A_\nu^{(\mu)} - \sum a_\nu^{(i)} a_k^{(\mu)} A_{\nu k}^{\mu i}. \quad (10)$$

Here  $i$  denotes every index different from  $\mu$ , and  $k$  every index different from  $\nu$ . The quantity  $A_{\nu k}^{\mu i}$  is at the same time the first subdeterminant, corresponding to the element  $a_k^{(i)}$ , of the  $(n - 1)$ -rowed determinant  $A_\nu^{(\mu)}$ .

For the bordered determinant

$$U = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \dots & a_n^{(1)} & u_1 \\ a_1^{(2)} & a_2^{(2)} & \dots & a_n^{(2)} & u_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_1^{(n)} & a_2^{(n)} & \dots & a_n^{(n)} & u_n \\ v_1 & v_2 & \dots & v_n & q \end{vmatrix} \quad (11)$$

one obtains, by developing with respect to the elements of the last row and column,

$$U = qA - \sum u_i v_k A_k^{(i)}. \quad (12)$$

In the higher subdeterminants it is sometimes useful to denote them by differential quotients; for example

$$A_{\nu k}^{\mu i} = \frac{\partial^2 A}{\partial a_{\nu}^{(\mu)} \partial a_k^{(i)}}. \quad (13)$$

## § 24. Linear homogeneous equations

The principal application of determinants, from which the whole theory took its origin, is the solution of linear equations. We treat the problem at once in its most general form, since the special form gives scarcely any simplification and is easily derived afterwards from the general result.

Consider a system of  $m$  equations of the first degree in which  $n$  unknowns  $x_1, x_2, \dots, x_n$  occur homogeneously:

$$\begin{aligned} a_1^{(1)} x_1 + a_2^{(1)} x_2 + \dots + a_n^{(1)} x_n &= 0, \\ a_1^{(2)} x_1 + a_2^{(2)} x_2 + \dots + a_n^{(2)} x_n &= 0, \\ &\vdots \\ a_1^{(m)} x_1 + a_2^{(m)} x_2 + \dots + a_n^{(m)} x_n &= 0. \end{aligned} \quad (1)$$

The coefficients  $a_i^{(k)}$  are regarded as given quantities. For the present we make no assumption about the numbers  $m, n$ , but ask generally for all systems of values of  $x_1, x_2, \dots, x_n$  satisfying (1).

One solution is immediate, whatever the coefficients may be:

$$x_1 = 0, x_2 = 0, \dots, x_n = 0. \quad (2)$$

There is also the other extreme case in which all coefficients vanish; then the equations are satisfied for arbitrary values of the unknowns.

We write the system of coefficients as the rectangular scheme

$$\begin{array}{cccc} a_1^{(1)} & a_2^{(1)} & \dots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \dots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(m)} & a_2^{(m)} & \dots & a_n^{(m)}. \end{array} \quad (3)$$

Such a scheme, which by itself has no numerical meaning, is called a matrix insofar as it is regarded as the source of a greater number of determinants. The determinants arising from the matrix are obtained by deleting arbitrary rows and columns, in such a number that the elements left form a square, and then regarding this square as a determinant.

Assume that among the  $\nu$ -rowed determinants of the matrix at least one is different from zero, while all  $(\nu + 1)$ -rowed and therefore all higher determinants, if such exist, vanish. Excluding the uninteresting case in which all coefficients vanish, such a number  $\nu$  always exists. Without loss of generality, we may suppose that the non-vanishing determinant is

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \dots & a_{\nu}^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \dots & a_{\nu}^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \dots & a_{\nu}^{(\nu)} \end{vmatrix}. \quad (4)$$

The subdeterminants of  $A$  are denoted, as before, by  $A_i^{(k)}$ , where  $i, k$  range from 1 to  $\nu$ .

I. If first  $\nu = n$ , which presupposes that  $m$  is not smaller than  $n$ , then the equations (1) have no other solution than (2).

For take the first  $n$  equations. If these are multiplied successively by  $A_\mu^{(1)}, A_\mu^{(2)}, \dots, A_\mu^{(n)}$ , where  $\mu$  is any of the indices  $1, 2, \dots, n$ , and then added, the relations of the preceding paragraph give

$$Ax_\mu = 0,$$

and since, by hypothesis,  $A \neq 0$ , it follows that  $x_\mu = 0$ .

II. In particular, if a system of  $n$  linear homogeneous equations with  $n$  unknowns has a determinant different from zero, all unknowns have the value zero. Equivalently: if such a system has a solution in which not all unknowns vanish, then the determinant of the system vanishes.

We now suppose that  $\nu < n$ . Since  $m \geq \nu$ , choose the first  $\nu$  equations and write them in the form

$$\begin{aligned} a_1^{(1)}x_1 + \dots + a_\nu^{(1)}x_\nu &= -a_{\nu+1}^{(1)}x_{\nu+1} - \dots - a_n^{(1)}x_n, \\ &\vdots \\ a_1^{(\nu)}x_1 + \dots + a_\nu^{(\nu)}x_\nu &= -a_{\nu+1}^{(\nu)}x_{\nu+1} - \dots - a_n^{(\nu)}x_n. \end{aligned} \quad (6)$$

If these equations are multiplied by the corresponding subdeterminants  $A_\mu^{(1)}, \dots, A_\mu^{(\nu)}$ , and added, one obtains

$$Ax_\mu = - \sum_{h=\nu+1}^n C_{\mu h} x_h, \quad (7)$$

where, for abbreviation,

$$C_{\mu h} = \sum_{i=1}^{\nu} a_h^{(i)} A_\mu^{(i)}, \quad h = \nu + 1, \dots, n. \quad (8)$$

By the theorem on subdeterminants,  $C_{\mu h}$  is the determinant obtained from (4) by replacing the  $\mu$ -th column by  $a_h^{(1)}, a_h^{(2)}, \dots, a_h^{(\nu)}$ . Thus (7), since  $A \neq 0$ , expresses  $x_1, \dots, x_\nu$  linearly in terms of  $x_{\nu+1}, \dots, x_n$  and the known coefficients.

III. The expressions (7) satisfy all equations (1), whatever values are assigned to  $x_{\nu+1}, \dots, x_n$ ; hence  $n - \nu$  of the unknowns remain arbitrary and the other  $\nu$  are dependent on them according to (7).

Indeed, writing (7) in the form

$$Ax_\mu = - \sum_{h=\nu+1}^n \sum_{i=1}^{\nu} x_h a_h^{(i)} A_\mu^{(i)}, \quad (9)$$

and multiplying by  $a_\mu^{(k)}$  and summing over  $\mu = 1, \dots, \nu$ , then adding the remaining terms, gives

$$A \sum_{\mu=1}^{\nu} a_\mu^{(k)} x_\mu = \sum_{h=\nu+1}^n x_h \left( A a_h^{(k)} - \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} \right). \quad (11)$$

The factor of  $x_h$  on the right is, by (12) of the preceding paragraph, a  $(\nu + 1)$ -rowed determinant of the matrix (3), and therefore vanishes by hypothesis; if  $k \leq \nu$  it vanishes because two rows coincide. Thus

$$\sum_{\mu=1}^{\nu} a_\mu^{(k)} x_\mu = 0, \quad k = 1, 2, \dots, m, \quad (13)$$

as was required.

A useful special case is  $m = n - 1$ ,  $\nu = n - 1$ . Then only one of the unknowns remains arbitrary, and the ratios of the unknowns are fully determined. If the  $(n - 1)$ -rowed determinants of the

matrix

$$\begin{array}{cccc} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(n-1)} & a_2^{(n-1)} & \cdots & a_n^{(n-1)} \end{array} \quad (14)$$

taken with alternating sign are denoted by  $A_1, A_2, \dots, A_n$ , and if at least one of them is different from zero, the solution of

$$\begin{aligned} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n &= 0, \\ &\vdots \\ a_1^{(n-1)}x_1 + a_2^{(n-1)}x_2 + \cdots + a_n^{(n-1)}x_n &= 0 \end{aligned} \quad (15)$$

is given by the ratios

$$x_1 : x_2 : \cdots : x_n = A_1 : A_2 : \cdots : A_n. \quad (16)$$

For  $n = 3$ , the system

$$ax + by + cz = 0, \quad a'x + b'y + c'z = 0 \quad (17)$$

has the solution in ratios

$$x : y : z = bc' - cb' : ca' - ac' : ab' - ba'. \quad (18)$$

## § 25. Elimination from linear equations

It sometimes happens that, for a given system of linear equations, the question is not so much the actual determination of the unknowns as the decision whether a solution is possible; that is, the setting up of the condition equations which must hold between the coefficients if solutions, or solutions of a prescribed kind, are to exist. The setting up of these condition equations is called elimination. The solution of this problem has already been contained implicitly in what precedes; nevertheless we return explicitly to a few questions belonging here.

Let again a system of  $m$  linear homogeneous equations with  $n$  unknowns be given, and ask when the system has a solution in which not all unknowns vanish. This is always the case when  $n > m$ . If, however,  $n \leq m$ , the necessary and sufficient condition for such a solution is that all  $n$ -rowed determinants of the matrix vanish. For if one of them does not vanish, the unknowns must all be zero by §24, II; but if they all vanish, one can find a number  $\nu < n$  such that all  $(\nu + 1)$ -rowed determinants vanish while at least one  $\nu$ -rowed determinant does not vanish, and then §24, III gives a solution of the required kind.

From the matrix one can form, when  $n \leq m$ ,

$$\frac{m(m-1) \cdots (m-n+1)}{1 \cdot 2 \cdots n}$$

$n$ -rowed determinants, and this would be the number of conditions. If  $n = m$ , this number is 1. In general, however, although all these conditions must be fulfilled, their number is greater than necessary, because some are necessary consequences of the others.

To obtain a system of necessary, sufficient, and mutually independent conditions, make the question more precise: what are the conditions that from a system of  $m$  linear homogeneous equations with  $n$  unknowns,  $\nu$  of the unknowns can be completely determined by the remaining  $n - \nu$ , which remain arbitrary?

Assume that  $x_{\nu+1}, x_{\nu+2}, \dots, x_n$  are to remain arbitrary and that  $x_1, x_2, \dots, x_\nu$  are to be determined by them, and assume

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix} \neq 0. \quad (1)$$

The unknowns  $x_1, \dots, x_\nu$  are then calculated by §24, (9), and one forms the sums §24, (11), whose vanishing expresses that the given system is actually satisfied. The conditions are therefore the vanishing of the determinants

$$\begin{vmatrix} a_1^{(1)} & \cdots & a_\nu^{(1)} & a_h^{(1)} \\ \vdots & & \vdots & \vdots \\ a_1^{(\nu)} & \cdots & a_\nu^{(\nu)} & a_h^{(\nu)} \\ a_1^{(k)} & \cdots & a_\nu^{(k)} & a_h^{(k)} \end{vmatrix} = 0, \quad (2)$$

or, written otherwise,

$$Aa_h^{(k)} - \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} = 0. \quad (3)$$

For  $h = \nu + 1, \dots, n$  and  $k = \nu + 1, \dots, m$  these give

$$(n - \nu)(m - \nu) \quad (5)$$

conditions. These conditions are mutually independent. For the left sides of (3) can, by suitable assumptions about the coefficients  $a$ , be made to take arbitrary values for each index combination, as is seen by setting all  $a_h^{(i)}, a_\mu^{(k)}$ , except the one in question, equal to zero.

## § 26. Non-homogeneous linear equations

We have simplified the problem of solving linear equations, and at the same time led it back to more general laws, by supposing the equations to be homogeneous in the unknowns. In applications, however, the unknowns often do not occur homogeneously. Although in principle the one case is not essentially different from the other, we consider the non-homogeneous case separately.

We can derive it from the homogeneous case by adding the requirement that one of the unknowns shall have the value 1. If this unknown belongs among those which the problem leaves arbitrary, there is no difficulty; if it belongs among those determined by the rest, further conditions have to be fulfilled. We treat the question independently, with a slightly changed notation, and restrict ourselves to the most important case in which the number of unknowns equals the number of equations.

Let the following system be solved for  $x_1, x_2, \dots, x_n$ :

$$\begin{aligned} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n &= y_1, \\ a_1^{(2)}x_1 + a_2^{(2)}x_2 + \cdots + a_n^{(2)}x_n &= y_2, \\ &\vdots \\ a_1^{(n)}x_1 + a_2^{(n)}x_2 + \cdots + a_n^{(n)}x_n &= y_n. \end{aligned} \quad (1)$$

The coefficients and the independent terms are regarded as given. Let

$$A = \sum \pm a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)} \quad (2)$$

be the determinant of the system, and let  $A_\mu^{(k)}$  denote its first subdeterminants. Multiplying the equations successively by  $A_\mu^{(1)}, A_\mu^{(2)}, \dots, A_\mu^{(n)}$  and adding, one obtains

$$Ax_\mu = \sum_{k=1}^n y_k A_\mu^{(k)}. \quad (4)$$

If  $A \neq 0$ , the values of the unknowns are uniquely determined and satisfy (1). If we put  $A_\mu^{(k)} = A_{k\mu}$ , the solution can be written in the form

$$\begin{aligned} Ax_1 &= A_{11}y_1 + A_{12}y_2 + \cdots + A_{1n}y_n, \\ Ax_2 &= A_{21}y_1 + A_{22}y_2 + \cdots + A_{2n}y_n, \\ &\vdots \\ Ax_n &= A_{n1}y_1 + A_{n2}y_2 + \cdots + A_{nn}y_n, \end{aligned} \tag{5}$$

which displays the analogy with the original system.

If  $A = 0$ , the equations (4) give no values for the  $x$ ; they only give conditions which the  $y$ 's must necessarily satisfy if the equations are to be soluble at all. Suppose that, besides  $A$ , all subdeterminants up to a certain order vanish, and denote by

$$B = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix} \tag{6}$$

a non-vanishing  $\nu$ -rowed subdeterminant. If all  $(\nu + 1)$ -rowed subdeterminants of  $A$  vanish, multiplication of the first  $\nu$  equations by the subdeterminants  $B_\mu^{(1)}, \dots, B_\mu^{(\nu)}$  gives

$$Bx_\mu = \sum_{i=1}^{\nu} B_\mu^{(i)} y_i - \sum_{h=\nu+1}^n x_h \sum_{i=1}^{\nu} B_\mu^{(i)} a_h^{(i)}. \tag{7}$$

Thus  $\nu$  of the unknowns are determined by the remaining ones. Substitution into (1) shows that the equations are satisfied if and only if, for  $\lambda = \nu + 1, \dots, n$ ,

$$By_\lambda = \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} B_\mu^{(i)} a_\mu^{(\lambda)} y_i. \tag{10}$$

These are  $n - \nu$  independent conditions for the  $y$ 's. If one of them is not fulfilled, the given system of equations has no solution.

If in (1) the  $x$ 's are regarded not as unknowns but as variables, the  $y$ 's also become variable quantities. In this interpretation the system (1) is called a linear substitution, since it mediates the passage from one system of variables to another. Only when the determinant  $A$ , now called the determinant of the substitution, is different from zero can the  $y$ 's be considered independent variables. In that case (5) gives the representation of the variables  $x$  by the variables  $y$ , or the substitution inverse to (1).